

May 2024
IB MAA HL P1 TZ1 Student C.

1. mean = 3

$$(a) \quad p + 2q + 3(4) + 4(2) + 6(3) = 3.$$

$$p + 2q + 38 = 3$$

$$\Rightarrow p + 2q = -35 \quad - (1)$$

$$p + q + 4 + 2 + 3 = 16$$

$$p + q = 7$$

— (2)

$$(1) - (2):$$

$$\Rightarrow q = -42, \quad p = 49$$

$$(b) \quad \text{mean final score} = 10(3)$$

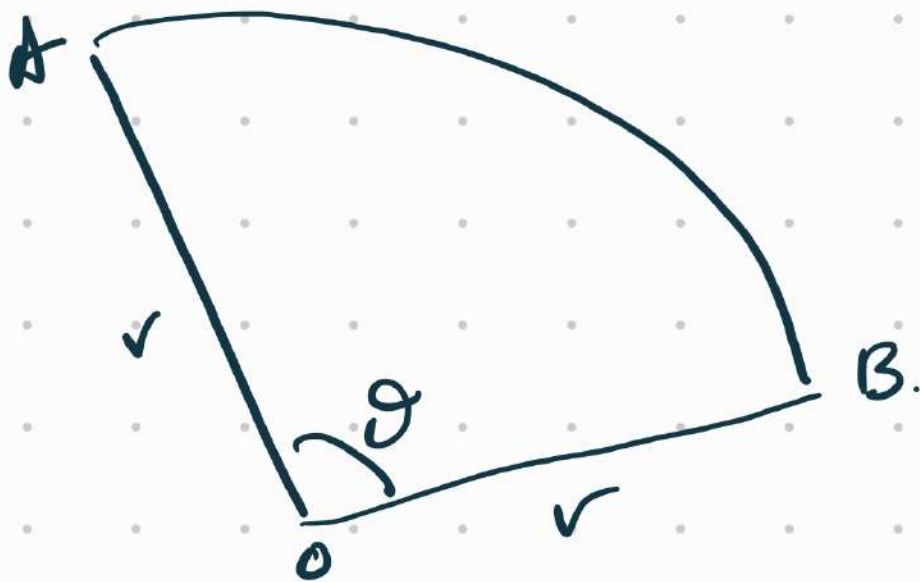
$$= 30.$$

$$2. \quad \log_{10} a = \frac{1}{3}, \quad a > 0$$

$$\log_{10} \left(\frac{1}{a} \right) = \frac{\log_{10} 1}{\log_{10} a} = \frac{0}{\frac{1}{3}} = 0$$

$$(b) \quad \log_{1000} a = \frac{\log_{10} a}{\log_{10} 1000} = 0$$

3.



$$(1) \text{ Perimeter} = r\theta = 10$$

$$\text{Area} = \frac{1}{2}r^2\theta = 6.25$$

$$\Rightarrow r^2\theta = 12.5$$

$$\Rightarrow \theta = \frac{12.5}{r^2} \quad - (2)$$

Sub. (2) into (1).

$$r\left(\frac{12.5}{r^2}\right) = 10$$

$$12.5r = 10$$

$$(b) \quad 4r^2 - 20r + 25 = 0$$

$$(2r - 5)(2r - 5) = 0$$

$$r = \frac{5}{2}$$

$$\theta = \frac{12 \cdot 5}{r^2} = \frac{12 \cdot 5}{\left(\frac{5}{2}\right)^2}$$

$$= \frac{12 \cdot 5}{\frac{25}{4}}$$

4. $f(x) = \cos x$, $g(x) = \sin 2x$
 $0 \leq x \leq \pi$.

(a) $\cos x = \sin 2x$.

$\Rightarrow \cos x = 2 \sin x \cos x$

$1 - 2 \sin x = 0$

or $\sin x = \frac{1}{2}$.

$x = \frac{\pi}{6}$ or $x =$

x -coordinate of A = $\frac{\pi}{6}$

x -coordinate of C =

(b) Area of R

$$= \int_{\frac{\pi}{2}}^{\pi} (\cos x - \sin 2x) dx$$

$$= \left[\sin x - \frac{\cos 2x}{2} \right]_{\frac{\pi}{2}}^{\pi}$$

$$= \left[\sin \pi - \frac{\cos 2\pi}{2} \right] - \left[\sin \frac{\pi}{2} - \frac{\cos \pi}{2} \right]$$

$$= -\frac{1}{2} - \left(1 + \frac{1}{2} \right)$$

$$= 2 \text{ units}^2$$

5. G.P. $u_1 = 1$, $r = 10$.

$$(a) S_n = \frac{1(10^n - 1)}{10 - 1} = \frac{10^n - 1}{9}$$

$$a = 10, b = 9$$

$$(b) S_1 + S_2 + S_3 + \dots + S_n$$

$$= \frac{10^1 - 1}{9} + \frac{10^2 - 1}{9} + \frac{10^3 - 1}{9} + \dots + \frac{10^n - 1}{9}$$

$$= \frac{1}{9} [10^1 + 10^2 + 10^3 + \dots - 1 - 1 - 1 - \dots]$$

$$= \frac{1}{9} [10 + 100 + 1000 + \dots + 10^n]$$

6. $f(x) = \sqrt{x \sin(x^2)}$, $0 \leq x \leq \sqrt{\pi}$.

Volume of revolution about x -axis

$$= \pi \int_0^{\sqrt{\pi}/2} x \sin(x^2) dx$$

$$= \pi \left[-\cos x^2 \right]_0^{\sqrt{\pi}/2}$$

$$= \pi \left[-\cos \frac{\pi}{4} + \cos 0 \right]$$

$$= -\pi \left(\frac{\sqrt{2}}{2} \right)$$

7. Prove by mathematical induction

$$\sum_{r=1}^n {}^r C_1 = {}^{n+1} C_2, \quad n \in \mathbb{Z}^+.$$

$$n=1 \quad \text{LHS} = {}^1 C_1 = 1 = \text{RHS}$$

True for $n=1$.

Let $n=k$

$$\sum_{r=1}^k {}^r C_1 = {}^{k+1} C_2$$

Let $n=k+1$

$$\begin{aligned} \sum_{r=1}^{k+1} {}^r C_1 &= \sum_{r=1}^k {}^r C_1 + {}^{k+1} C_1 \\ &= {}^{k+1} C_2 + {}^{k+1} C_1 \end{aligned}$$

$$= \frac{(k+1)!}{2! (k+1-2)!} + \frac{(k+1)!}{1! (k+1-1)!}$$

$$= \frac{(k+1)!}{2! (k-1)!} + \frac{(k+1)!}{k!} = {}^{k+2} C_2$$

Since it is true for $n=1$, and
true for $n=k \Rightarrow n=k+1$ is true,
by mathematical induction,

$$\sum_{r=1}^n r C_1 = {}^{n+1}C_2, \quad n \in \mathbb{Z}^+$$

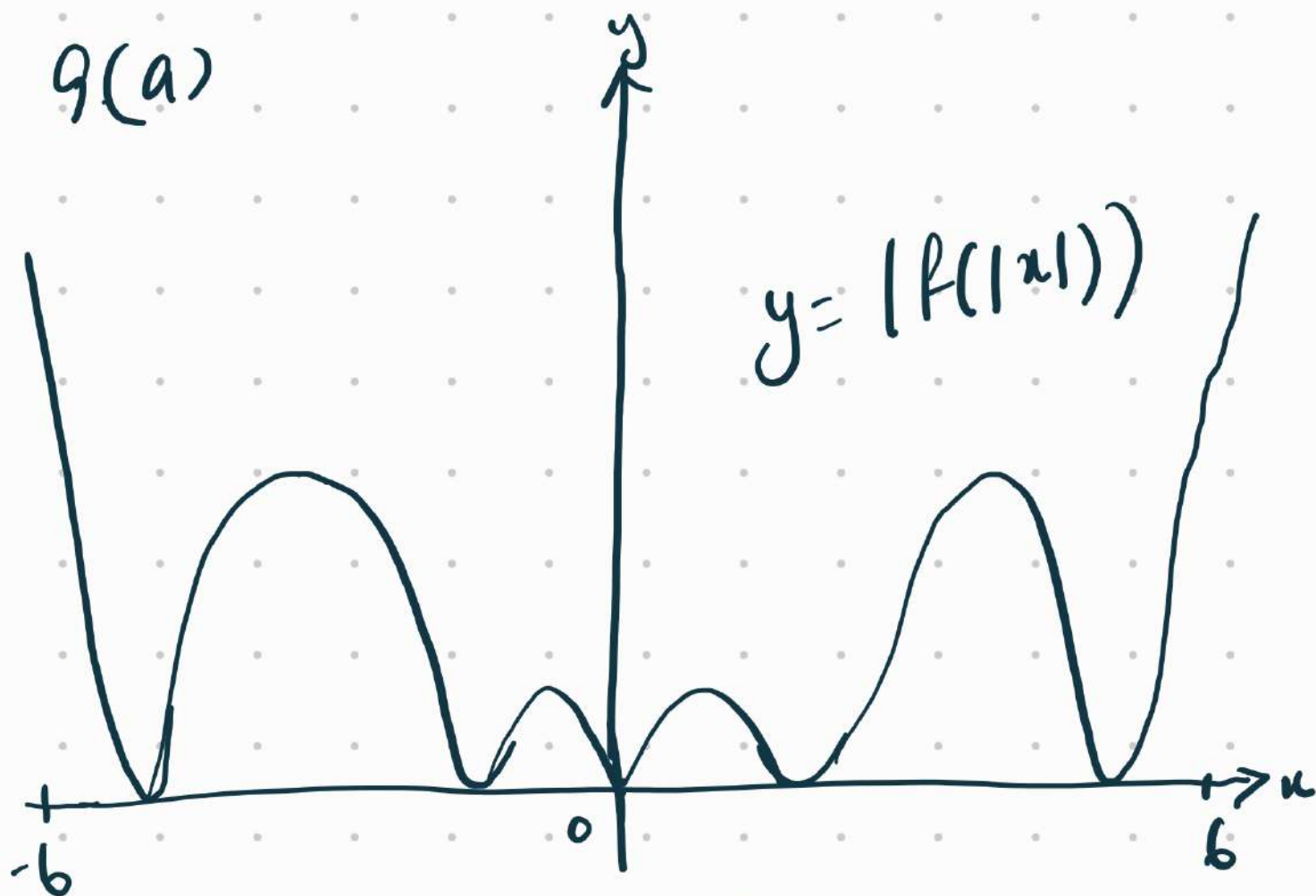
8. (a) Maclaurin series:

$$(i) \sin(x^2) = x^2 - \frac{(x^2)^3}{3!} \\ = x^2 - \frac{x^6}{6}$$

$$(ii) \sin^2(x^2) = \left(x^2 - \frac{x^6}{6} + \dots\right) \left(x^2 - \frac{x^6}{6} + \dots\right) \\ = x^4 - \frac{x^8}{6} - \frac{x^8}{6} \\ = x^4 - \frac{x^8}{3}$$

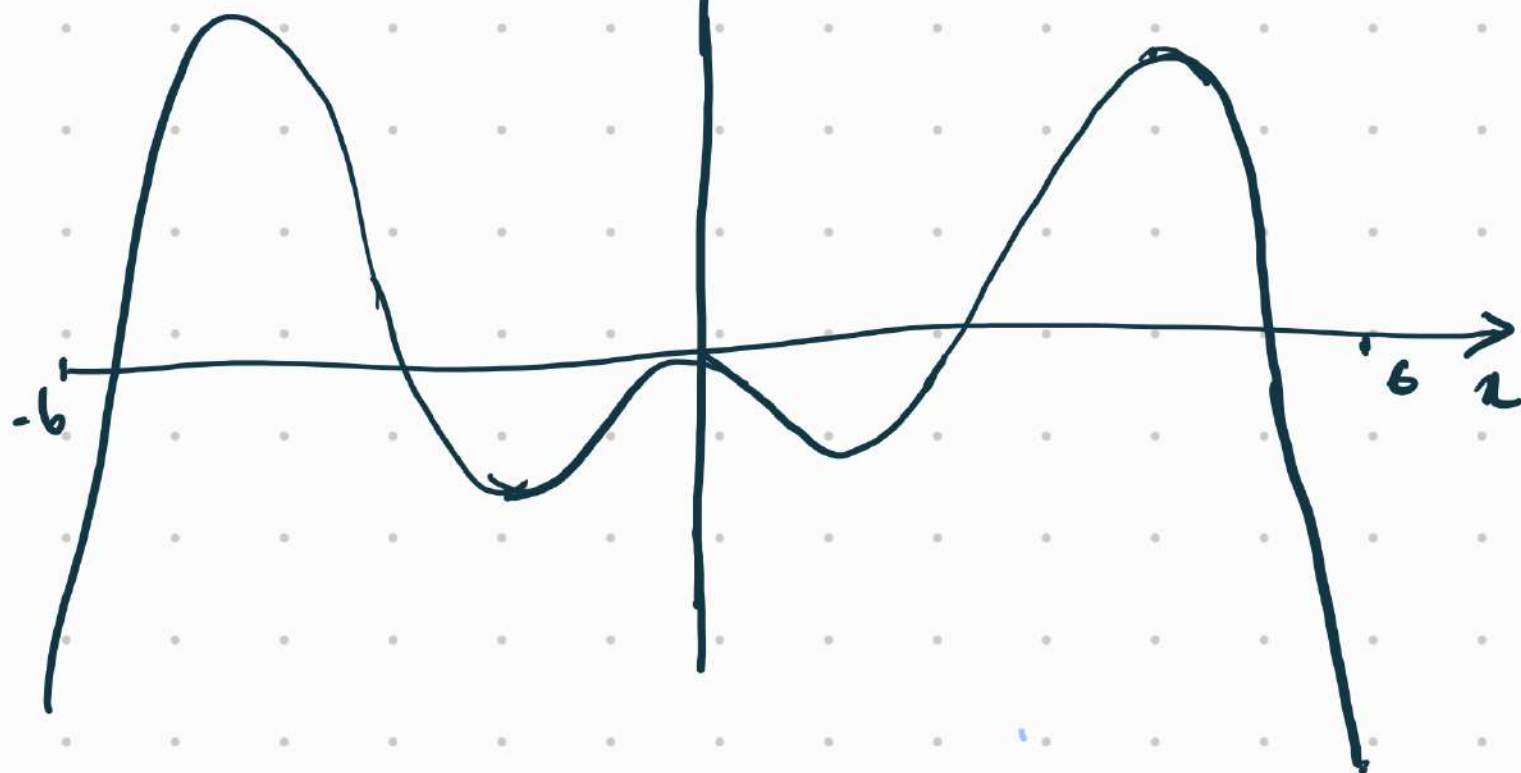
$$(b) 4x \sin(x^2) \cos(x^2) \\ = 4x \left(x^2 - \frac{x^6}{6}\right) \left(1 - \frac{x^2}{2!}\right) \\ = 4x \left(x^2 - \frac{x^6}{6} - \frac{x^4}{2} - \frac{x^8}{12}\right) \\ = 4x^3 - 2x^5$$

$g(a)$



y

$y = f(x)$



$$(c) \int_0^4 f(|x|) dx = 1.6$$

$$(i) \int_{-4}^0 f(x) dx = 1.6$$

$$(ii) \int_{-4}^4 f(|x|) + f(x) dx$$
$$= \int_{-4}^4 f(|x|) dx + \int_{-4}^4 f(x) dx$$

$$= 3.2 + 3.2$$

$$= 6.4$$

10. $f(x) = \frac{4x+2}{x-2}, x \neq 2.$

(a) $= 4 + \frac{10}{x-2}$

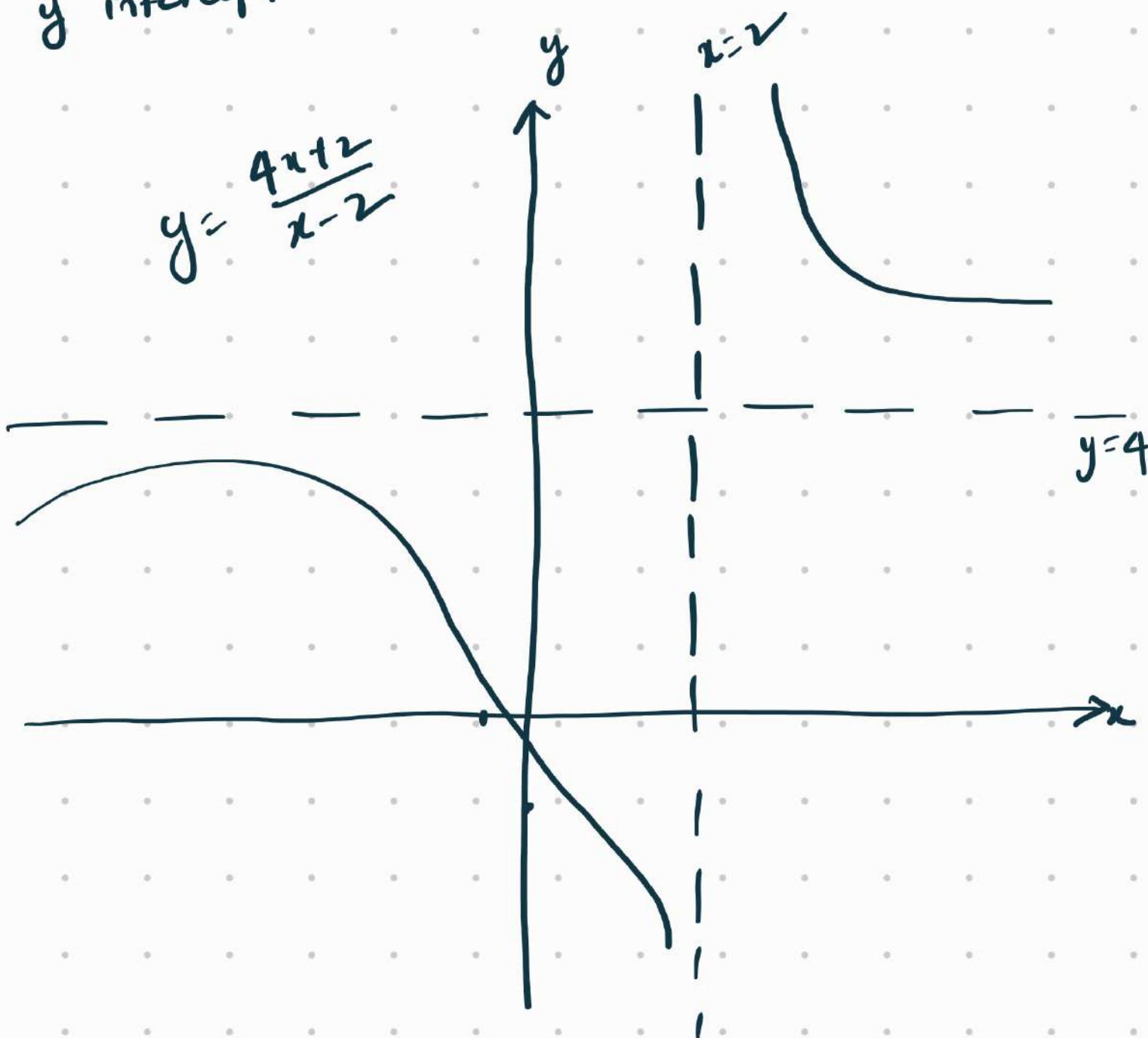
vertical asymptote: $x=2.$

horizontal asymptote: $y=4.$

x intercept: when $y=0, x=-\frac{1}{2}$

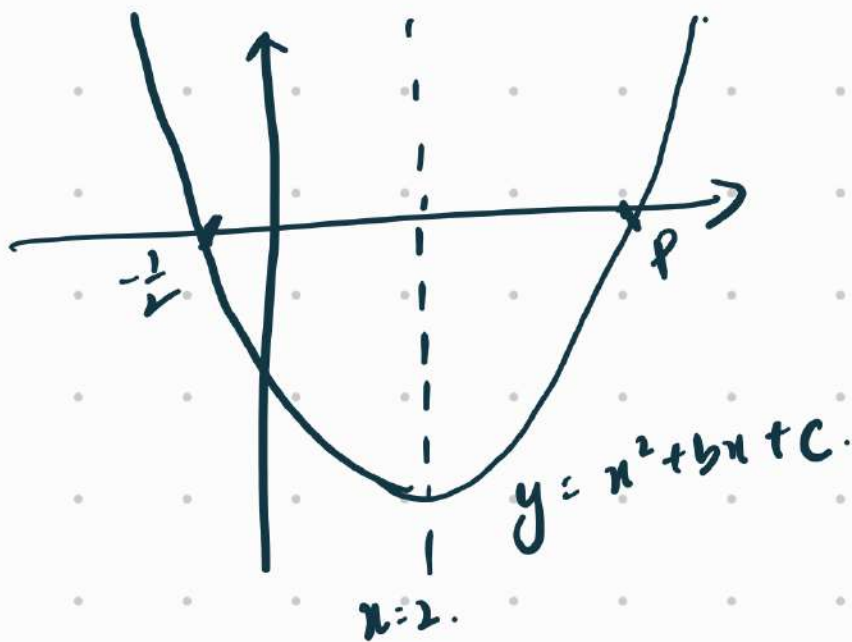
y intercept: when $x=0, y=-1$

$$y = \frac{4x+2}{x-2}$$



(b) Range of f .
 $y \neq 4$.

(c) $g(x) = x^2 + bx + c$
Axis of symmetry: $x = 2$.



$$\frac{p - \frac{1}{2}}{2} = 2 \Rightarrow p = 4.5 = \frac{9}{2}.$$

(d) $x^2 + bx + c = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4c}}{2} = 0$$

$$\frac{-b - \sqrt{b^2 - 4c}}{2} = -\frac{1}{2}$$

$$b + \sqrt{b^2 - 4c} = 1 \quad -(1)$$

$$\frac{-b + \sqrt{b^2 - 4c}}{2} = \frac{9}{2}$$

$$-b + \sqrt{b^2 - 4c} = 9 \quad -(2)$$

$$\begin{aligned} (e) \quad g(2) &= 2^2 + 2b + c \\ &= 4 + 2b + c \end{aligned}$$

$$y\text{-coordinate of vertex} = 4 + 2b + c$$

(f) let $b=1, c=2$.

$$f(x) = g(x)$$

$$\frac{4x+2}{x-2} = x^2 + x + 2$$

$$4x+2 = (x^2 + x + 2)(x-2)$$

$$4x+2 = x^3 - 2x^2 + x^2 - 2x + 2x - 4$$

$$x^3 - x^2 - 4x - 6 = 0$$

$$\text{product of roots} = 6$$

$$11. \quad P(x) = 3x^3 + 5x^2 + x - 1$$

$$(a) \quad P(-1) = 3(-1) + 5(1) - 1 - 1 \\ = 0$$

Hence, $x+1$ is a factor of $P(x)$

(b)

$$P(x) = (x+1)(3x^2 + 2x - 1)$$

$$= (x+1)(3x-1)(x+1)$$

$$= (3x-1)(x+1)^2$$

$$\begin{array}{r} x+1 \overline{) 3x^3 + 5x^2 + x - 1} \\ \underline{3x^2} \\ 2x^2 + x - 1 \\ \underline{2x^2 + 2x} \\ -x - 1 \\ \underline{-x - 1} \\ 0 \end{array}$$

$$Q(x) = (x+1)(2x+1)$$

$$(c) \quad \frac{1}{Q(x)} = \frac{1}{(x+1)(2x+1)}$$

$$= \frac{1}{x+1} + \frac{1}{2x+1}$$

$$\begin{aligned}
 (d) \quad & \frac{1}{(x+1)Q(x)} \\
 &= \left(\frac{1}{x+1} \right) \left[\frac{1}{x+1} + \frac{1}{2x+1} \right] \\
 &= -\frac{1}{(x+1)^2} + \frac{1}{(x+1)(2x+1)} \\
 &= -\frac{1}{(x+1)^2} + \frac{1}{x+1} + \frac{1}{2x+1}
 \end{aligned}$$

$$\begin{aligned}
 (e) \quad & \int \frac{1}{(x+1)^2(2x+1)} dx \\
 &= \int -\frac{1}{(x+1)^2} + \frac{1}{x+1} + \frac{1}{2x+1} dx \\
 &= (x+1)^{-1} + \ln(x+1) + \ln(2x+1)
 \end{aligned}$$

12. $\phi = (a+bi)^3$, $a, b \in \mathbb{R}$.

$$\begin{aligned} \text{(a)} \quad (a+bi)^3 &= (a+bi)(a+bi)(a+bi) \\ &= (a^2 + 2abi + b^2)(a+bi) \\ &= a^3 + a^2bi + 2a^2bi + 2ab^2 + b^2a + b^3i \\ &= (a^3 + 3ab^2) + (3a^2b + b^3)i \end{aligned}$$

(i) $\operatorname{Re}(\phi) = a^3 + 3ab^2$

(ii) $\operatorname{Im}(\phi) = 3a^2b + b^3$

(b) $(1 + \sqrt{3}i)^3$

$$= [1 + 3(3)] + [3\sqrt{3} + (\sqrt{3})^3]i$$

$$= 10 + [3\sqrt{3} + (\sqrt{3})^3]i$$

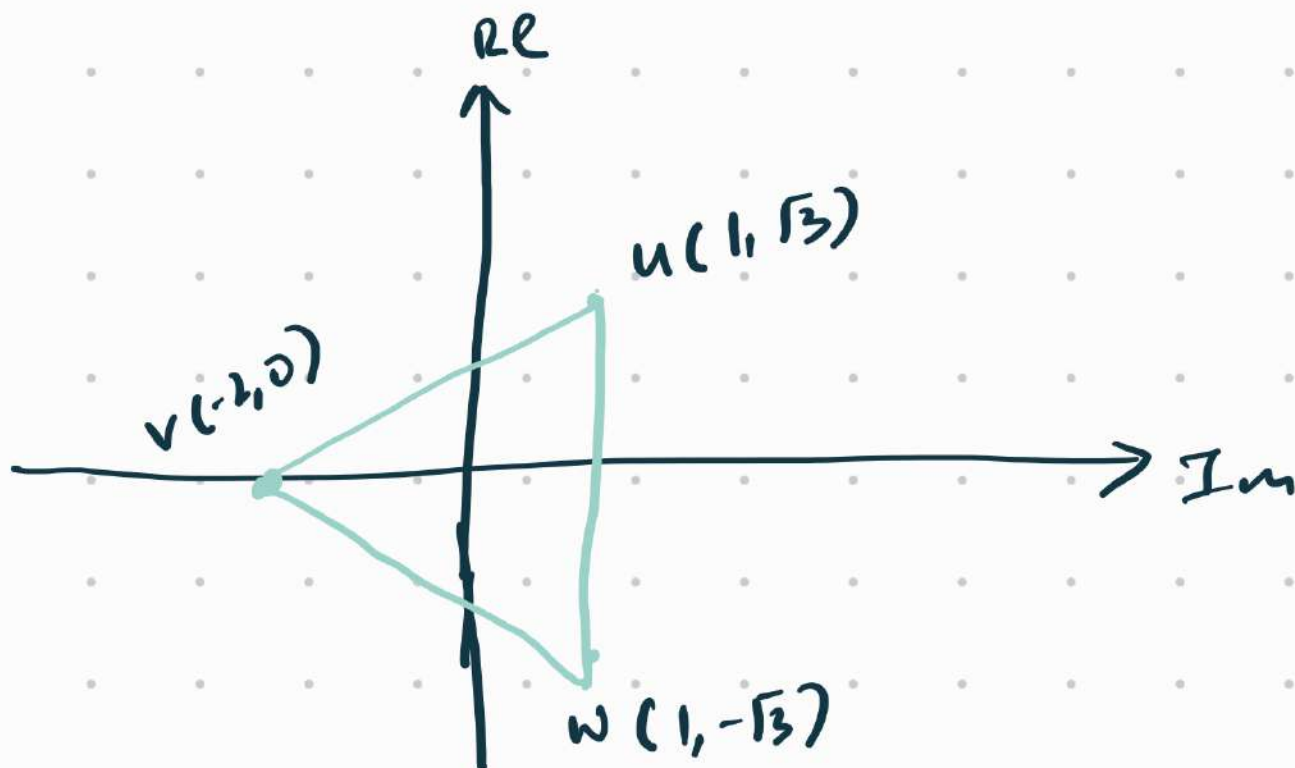
$$z^3 = -8,$$

$$u = 1 + \sqrt{3}i \quad v \in \mathbb{R}.$$

(c) $v = -2.$

$$w = 1 - \sqrt{3}i \quad (\text{conjugate})$$

(d)



$$\begin{aligned} \text{Area of } \triangle uvw &= \frac{1}{2} (2\sqrt{3}) (3) \\ &= 3\sqrt{3} \text{ units}^2 \end{aligned}$$

(e) $u = 1 + \sqrt{3}i = 2e^{i(\frac{\pi}{3})}$

$$v = -2 = 2e^{i\pi}$$

$$w = 1 - \sqrt{3}i = 2e^{i(\frac{5\pi}{3})}$$

(f) u', v', w' are solutions of
 $z^3 = c + di$

(g) $n = 10$